

Outage prediction models for snow and ice storms

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ABSTRACT

This paper describes the development of two Outage Prediction Models (OPMs) for power outages caused by snow and ice storms on electric distribution networks, and their performance evaluation in the Northeastern United States. The first is a Machine Learning (ML) based model that is set up to predict power outages on a regular grid (4-km grid spacing). The second is a Generalized Linear Model (GLM) that is set up to predict total power outages at the town level. Both models are fed with Numerical Weather Prediction (NWP) outputs, satellite-derived leaf area index (LAI) and land cover data, and utility specified infrastructure and historical outage data. For both models the most important variables are the amount of assets on the territory, LAI, snow density, and – for the ice model – the amount of freezing rain. Results from cross-validation experiments based on 54 outage events spanning three orders of magnitude show median absolute percentage errors around 70% for both models. The GLM is able to predict extreme events (such as the devastating October 2011 nor'easter) better than ML models do, while ML models have better performance for lower impact events and present better characterization of the spatial distribution of power outages.

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1. Introduction

Atmospheric icing is the accretion of precipitation, in-cloud droplets, or hoarfrost on the surface of an object [1,2]. In recent years, a large number of widespread power outage events have resulted from extreme atmospheric icing on electricity transmission and distribution networks caused by ice storms (including Wisconsin in 1976 [3]; Canada and the northeastern United States in 1998 [4,5]; North Carolina in 2002 [6]; the Great Plains in 2007 [7]; and Hunan, China in 2008 [8,9]) and wet snow storms (including Germany in 2005 [10,11]; Spain in 2010 [12]; and the Northeastern United States in 2011 [13] and 2018).

The costs from such outage events have been enormous. In the United States, winter storm losses between 1988 and 1995 averaged \$375 million (in 1997 U.S. dollars) per year, of which 60% were caused by ice storms [14], while the economic damage from the 1998 ice storm in Canada was much higher. With 4.7 million Canadians left without power, Hydro-Quebec and Ontario Hydro incurred repair costs estimated at 1 billion in Canadian dollars (CAD), while short-term losses in manufacturing, transportation, communication, and retail sales reached 1.6 billion CAD, and income loss amounted to 1 billion CAD [4]. In October 2011, a wet snowstorm in the northeastern United States left 3.5

million people without power [15], causing damages estimated between 1 billion and 3 billion USD [16]. In Connecticut, costs for restoring power to 830,000 customers of CL&P (now Eversource Connecticut) amounted to 175 million USD. More recently, in 2018, four powerful snowstorms hit the Northeastern United States, each producing wet snow and ice resulting in several hundreds of millions of dollars in damage and power outages lasting for several days.

Beyond economic losses, snow and ice storms causing widespread power outages raise health issues. Outbreaks of carbon monoxide poisoning, produced mainly by the operation of charcoal grills and electric generators in the absence of power, occurred in Washington State between 1993 and 1996 [17], after the 1998 ice storm in Maine [18], after the 2002 ice storm in North Carolina [19], and after the 2011 snow storm in Connecticut [20].

Predicting atmospheric icing and its effects on electric distribution networks is therefore necessary to evaluate and adopt preventive measures and reduce calamitous losses like the ones discussed above [21]. To this end, several physical and empirical approaches have been used to model atmospheric icing (in the form of ice and snow accretion) on structures [22,23].

Among them have been analytical and physical models for estimating ice accretion based on: (i) heat transfer (wind and temperature) by neglecting precipitation intensity [24]; (ii) precipitation intensity alone by neglecting effects of wind and temperature [25]; (iii) combined effects of wind speed and precipitation rate [26]; and (iv) wind and fall speed of the rain drops,

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and precipitation amount during the accretion time [27,28]. Also in use are numerical ice thickness models: i.e. models based on wind speed, temperature, pressure, cloud liquid water content, droplet size distribution and wire size [29]; and on temperature, wind speed, droplet diameter and liquid water content [30], an improved version of which is able to take into account cable diameter, angle between wind and line orientation, event duration, and formation of icicles [31].

Snow accretion on power lines has been also widely investigated. According to [32], in dry conditions, snow accretion is limited to low- and medium voltage lines shielded against wind (less than 2 m s^{-1}), while in wet conditions heavy loads on power lines are associated with high precipitation intensities and above freezing temperatures. Wet snow accretion models [10,33–35] incorporate accretion intensity as proportional to wind speed, sticking efficiency (proportional to wind speed [35]), snow terminal velocity, and concentration of wet snow in air (proportional to precipitation intensity [35], or to visibility [10]).

In the past two decades, researchers have begun to apply machine learning techniques to the prediction of ice accretion on overhead conductors. These have included the use of neural networks [36] trained on temperature, wind speed and direction, precipitation rate [22,37] and humidity ([22] only). These studies have concluded that such models do not give acceptable prediction accuracies [37], and that 'precise point-by-point forecast seem[ed] unreasonable' [22]. More promising results were found with a Support Vector Machine (SVM, [38] for learning ice accretion patterns using wind speed, wet bulb temperature, precipitation and fraction of frozen precipitation in eight icing events simulated based on a Numerical Weather Prediction (NWP) model [23].

All these models predict icing accretion on power lines, but they do not predict the damage it may cause on the electric distribution network. A study that took a step further [39] employed a generalized linear mixed model (GLMM, [40]) to estimate the number of power outages in eight ice storms through the use of number of protective devices, ice thickness (from NWP simulations, using [26]'s ice accretion model) and ice storm indicator. Another advance in predictive modeling of snow impact was the use of more than fifteen explanatory variables to predict snow-related power outages through tree-based models [41, 42]. These ice and snow impact models belong to the broader category of storm outage prediction models (OPMs) developed for hurricanes [39,41–52], extratropical storms [41,42,53], and thunderstorms [41,42,53].

In outage prediction models, errors arise from both weather forecasting inaccuracies and limitations of statistical models [53]. With respect to snow and ice storms additional uncertainties arise from inaccurate prediction of precipitation type, primarily freezing rain and ice pellets [54–62]. An incorrect precipitation type prediction is magnified by an outage model for winter storms, since the impact of freezing rain in terms of power outages may be orders of magnitude higher than the impact of ice pellets, which do not accumulate on power lines.

In this study, we investigated the challenging task of snow and ice storm outage prediction through the use of statistical and machine learning techniques, and we identified the primary factors responsible for winter storms outage prediction inaccuracies. Building upon existing literature, we created the first OPM entirely dedicated to snow and ice storms, able to classify weather conditions at a fine scale, and trained with a number of events representative of varying wintry conditions. In the following, a description of the study area and the methods for selecting and simulating the 54 snow and ice storms in our dataset is followed by an examination of the relationships among the snow- and ice-related variables for designing and validating two systems

capable of predicting snow- and ice-related power outages. The two systems use variables quantifying the amount of leaves on trees, freezing rain, snow and other precipitation types, soil moisture, winds and gusts during storms, matching this information with soil type, elevation, land cover and infrastructure. Each system is cross-validated in different configurations using the entire dataset of snow and ice storms to find the optimal structure, based on the overall OPM performance evaluation. Prediction performance of the two systems are presented as errors between predicted and actual outages for each storm, and spatial maps of errors, in order to infer conclusions on the usefulness of each model for snow and ice storm preparedness by utility managers. A particular discussion of the inaccuracies in weather and outage predictions for the October 2011 nor'easter is also presented.

2. Datasets and study area

The outage prediction models developed through this work are based on variables related to weather, infrastructure, land cover, elevation and vegetation. The methods used for obtaining the variables composing our dataset are described in this section, while a detailed description of each input variable can be found in [Appendix B](#).

2.1. Response variable

The response variable for the outage prediction models is the number of outages occurring over a certain area in a given storm. Electric utilities often keep records of power outages, and Eversource Energy-Connecticut (serving 1.2 million electric customers) provided us with an outage dataset containing geolocated power outages that occurred between 2005 and 2018 in Connecticut. Each outage is defined as a location where a two-man crew is needed to restore power, and may affect up to several hundreds customers.

2.2. Storm list

The outage dataset represents the basis for the storm selection. Following the criteria described in [53], we identified snow and ice events: (i) we identified the 95th percentile of the daily outage distribution and we selected the storms covering high outage days that resulted in at least 5 mm water equivalent of wintry precipitation over 25 percent of the territory; (ii) we added to this list storms associated with at least 10 mm of wintry precipitation measured at one or more airport stations, but not associated to widespread outages. This second criterion is needed because we do not know a priori whether a storm would cause outages or not, hence the training dataset needs to include both low and high impact winter storms. Storms that did not produce significant outages and were associated with low amounts of wintry precipitation were not included in the winter model. However, these storms were included in the extratropical storms model described in [53] if they were associated with strong winds. Through these principles, we selected 54 snow and ice events (listed in [Table A.1](#)) that occurred in Connecticut between 2005 and 2018.

2.3. Predictor variables

Predictor variables describe weather and leaf conditions during storms, land cover, elevation and soil characteristics of the territory; and infrastructure.

Weather variables have been derived from storm-based simulations using the Weather Research and Forecasting (WRF-ARW

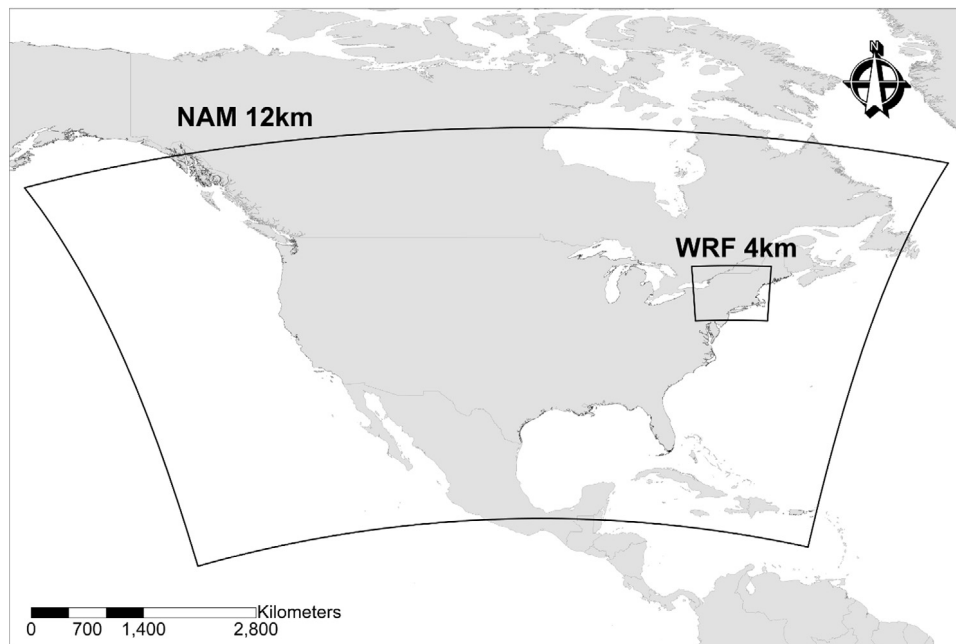


Fig. 1. NAM 12 km domain, covering the entire CONUS, and WRF domain, centered on the Northeastern United States, with a 4 km grid spacing.

Table 1

Parametrization schemes used in the WRF model.

Parametrization	Scheme
Microphysics	New Thompson et al. [67]
Cumulus	Grell 3D Ensemble Scheme [68,69]
Planetary Boundary Layer	Yonsei University [70]
Surface Layer	MM5 Similarity [71–75]
Land Surface Model	Unified Noah [76]
Longwave Radiation	RRTMG [77]
Shortwave Radiation	RRTMG [77]

v.3.8.1 [63,64]) NWP model. The WRF simulations run over a domain with 4 km horizontal grid spacing covering the Northeastern United States (Fig. 1). We used the North American Mesoscale (NAM) Forecast System [65] analysis at 12 km grid spacing as initial and boundary conditions.

The WRF model configuration is described in Table 1. Each storm simulation started 12 h before the beginning of the storm, lasting for 60 h. A 12 h spin-up time was then discarded, to obtain 48 h of useful storm analysis. For each of the 803 4-km resolution grid cells over Connecticut, weather conditions occurred during each storm were summarized into maximum, mean, and occurrence variables, reported in Appendix B. Among these variables, particular attention was given to the Air Force Weather Agency (AFWA) precipitation type variables, [66], such as snow, ice pellets, and freezing rain, since the focus of this study is to create a model for predicting outages in wintry conditions.

Infrastructure data are represented by the sum of isolating devices in each 4-km grid cell of the WRF model, since outages are usually reported to the closest isolating device. Land cover data (developed area, deciduous and coniferous forest) were derived from the National Land Cover Database (NLCD 2011, [78]). Similarly to [41,42,51,52] and [53], data were aggregated into the WRF grid cells by considering only the land cover in proximity of overhead lines. Elevation was computed, for each cell, as the mean elevation of the overhead lines, from the USGS 3DEP Digital Elevation Model (DEM) at 1 arc second resolution [79]. Soil data (hydric, sandy, rocky, depth of rocks) were derived from the USGS Soil Survey Geographic Database (SSURGO, [80]). Leaf Area

Index (LAI) data were obtained from [53], who created a weekly climatological product by processing MODIS [81] observations.

Beyond the variables mentioned above, we derived three new variables, from the WRF output, to model power outages associated with snow: *snow_density*, *snow_time* and *LAI_snow*.

Snow_density is a fundamental variable for discerning wet snow from dry snow events. Since the AFWA module provides both snow accumulation and snow water equivalent, it is possible to estimate snow density by computing the ratio between the snow water equivalent and snow accumulation.

Many icing accretion models are based on snow amount and wind speed. Since the weather variables extracted from WRF do not provide any information about the co-occurrence of wind and snow, a new variable, *snow_time*, was introduced as the ratio between the amount of snow fallen during the four hours of maximum wind, and the total amount of snow.

Lastly, since most of the power outages in the study area (Northeast United States) are caused by the interaction between vegetation and overhead lines, we introduced a quantity, *LAI_snow*, describing the combined effect of snow and vegetation amount, through the product of snow amount with LAI.

2.4. Snow and ice data analysis

In this section we analyze the winter storm dataset, with particular attention to understanding the role snow density may have on power outages. The 803 grid cells for each of the 54 winter storms analyzed in this study represent a sample large enough (43,362 entries) to infer important characteristics of the distribution of mixed-phase precipitation variables in Connecticut.

Histograms of (a) snow, (b) freezing rain, and (c) sleet (comprising here both graupel and ice pellets) for varying snow density (light, average and wet snow, delimited by red, vertical lines) are shown in Fig. 2. The value of 66.7 kg/m³ corresponding to a snow ratio of 15:1, is taken as reference point for the limit between light and average snow density [82], while the value of 105 kg/m³ (limit between average and wet snow) slightly differs from the 9:1 ratio (111.1 kg/m³) used in [82] due to technical reasons described in the next paragraphs. Dark bars in the histograms

represent data associated with snow total amounts above 2.5 mm water equivalent (moderate snow, [83]).

From the histogram it is evident that most of the mixed-phase precipitation data for Connecticut correspond to average snow densities. Moderate amounts of dry snow never manifest (dry snow occurs for very low temperatures), freezing rain is never associated with moderate amounts of wet snow (freezing rain is associated with a low-level inversion, while wet snow with an isothermal profile), and sleet has a bimodal distribution.

The reason why the limit between average and wet snow has been set to 105 kg/m³ is evident from Fig. 3 a, b, c, where the mean number of outages per 100 assets is plotted as a function of snow density. For snow densities below 105 kg/m³ we notice an increasing, linear trend of mean outages for increasing snow density (Fig. 3 a, b). Above this threshold, we note a sharp increase of outages in all storms, the most notable is for the 2011 nor'easter, which produced heavy snow over almost the entire service territory. In all cases, the peak of mean outages is around 115 kg/m³, and lower values can be found below 110 kg/m³ and above 120 kg/m³.

There is therefore a well defined range of snow density values where storms can create large amounts of outages, and this range is between 110 kg/m³ and 120 kg/m³, which is associated with temperatures between 1 °C and 2 °C (not shown). This result is consistent with [32], who found heavy snow loads associated with positive temperatures, due to capillary forces of liquid water and metamorphosis of snowflakes into ice, processes that are negligible or absent at lower temperature. When temperatures are even higher, partially melted falling snow does not stick on power lines because either it is not able to refreeze or it melts quickly. This behavior reflects in a sharp decrease of mean outages for snow densities above 120 kg/m³.

The effect of freezing rain on power lines is outlined in Fig. 3d: no significant trend is evident for values below 20 mm (corresponding to 8 mm of radial ice accumulation, following [84]), while a strong outage increase is present for higher accumulation. The value of 8 mm of radial ice accumulation is not far from the threshold of 0.25 inches, used by the United States National Weather Service as threshold for an experimental product for ice storm damage potential [85].

3. Methodology

In this section, after presenting the OPM system for snow and ice storms based on machine learning models and the one based on a generalized linear model, we introduce a grid cell classification for precipitation type. Such classification builds on the results of Section 2.4 to guide machine learning models and direct them towards performance improvements based on statistical findings or knowledge of the weather model uncertainties. Finally we describe the methodologies and the error metrics employed for validation and performance evaluation of the models.

3.1. Machine learning models

We used the two best performing machine learning models utilized in [53], following the same configuration:

- a Random Forest (RF) model [86], formed by 200 decision trees. Each decision tree is able to split the dataset into subsets for which the response variable is similar, by minimizing the sum of the square error [87]. Each tree of the random forest is trained on a random subset of training data using a random subset of explanatory variables. The R package 'randomForest', based on [86], is used for this purpose. In our configuration, hyperparameters other than the number of trees assume default values;

- a Bayesian Additive Regression Tree (BART, [88]) model, obtained through statistical sum of thirty Bayesian trees, that are the base learners. A Markov Chain Monte Carlo (MCMC, [89]) algorithm is used with ten thousand iterations for fitting the BART model, and four thousand iterations are used for obtaining the predictions. The R package 'BART', based on [88], is used to run the model.

RF and BART were combined through the optimization module described in [53]. The optimization consisted in the coefficient selection for the linear combination of the two models, by maximizing the ratio between the Nash–Sutcliffe Efficiency (NSE; [90]) and the median absolute percentage error (MdAPE). By combining five ML models through this procedure, in [53] we found a 50% reduction in the mean absolute percentage error (MAPE) with respect to the best performing model. For the current study the MAPE reduction deriving from the combination of BART and RF is 20%.

3.2. Generalized linear model

In addition to the machine learning models, a simpler Generalized Linear Model (GLM; [91,92]) was tested for predicting power outages. GLMs represent an extension of linear models for non Normally distributed variables. In a GLM the response variable Y_i is expressed as a linear function of multiple predictors plus an error ϵ_i :

$$Y_i = f(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_n x_{ni}) + \epsilon_i = f(\eta_i) + \epsilon_i \quad (1)$$

where $\beta_0 \dots \beta_n$ are coefficients, $f(\dots)$ is the inverse of the link function $g(\mu_i) = \eta_i$ that describes the dependency of the mean $E(Y_i) = \mu_i$ on the linear predictor, and a variance function $\text{var}(Y_i) = \phi V(\mu_i)$ describes how the variance depends on the mean, with $\phi = \text{const}$. Under the Poisson assumption, used for modeling the number of power outages at the town level, $Y_i = P(\lambda_i)$ where $P(\lambda_i)$ is the Poisson distribution, then $E(Y_i) = \lambda_i$, $\text{var}(Y_i) = \lambda_i$ hence the variance function is $V(\mu_i) = \mu_i$, and the canonical link function is $g(\mu_i) = \ln(\mu_i)$.

A more complex model, the spatial generalized linear mixed model (GLMM), was used in [39] for power outage prediction. However, we opted for the simpler GLM because in [39] the authors concluded that, although being some input variables spatially correlated, models simpler than the spatial GLMM can produce similar performance results.

3.3. Grid cell classification

In view of the uncertainties in precipitation type prediction and of the effects that different precipitation types have on power outages, we split the dataset into three categories for model training purposes: (i) wet snow, (ii) freezing rain, ice pellets and graupel, and (iii) rain, wind and dry snow, based on the decision tree shown in Fig. 4.

The algorithm developed for precipitation type classification at a grid cell level follows the rules listed below:

- grid cells with snow density above 105 kg/m³ are isolated from the rest;
- for such grid cells, it is evaluated if a moderate amount of snow fell during the storm and if that amount was greater than the amount of precipitation fallen as rain. Under these conditions, a cell is classified as heavy, wet snow, otherwise as rain, wind and dry snow.
- for the grid cells that are not associated with high snow density, it is evaluated if a moderate amount of solid precipitation different than snow occurred. Under this condition, a cell is classified as ice, otherwise as rain, wind and dry snow.

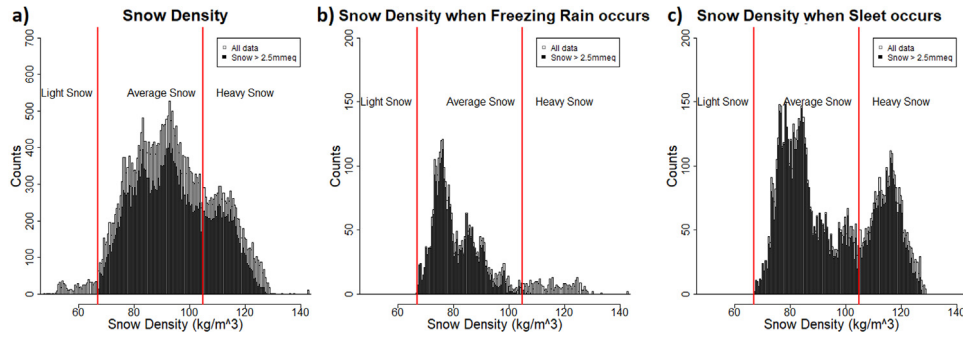


Fig. 2. Histograms representing the number of records in the dataset with snow (a), freezing rain (b) and sleet (c) for varying snow density. Dark colors correspond to snow amounts above 2.5 mm w.e., light colors to snow amounts below 2.5 mm w.e. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

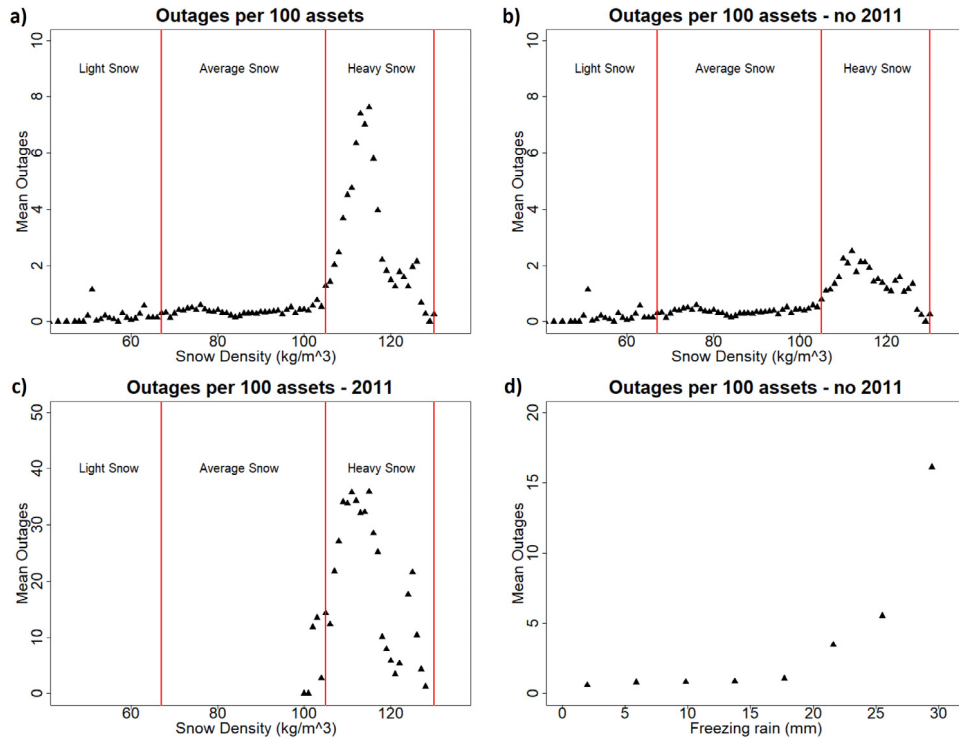


Fig. 3. Mean number of outages per 100 assets by snow density for: (a) the entire dataset, (b) all the storms except for the 2011 nor'easter, (c) the 2011 nor'easter; (d) mean number of outages per 100 assets by freezing rain amount for all the storms except for the 2011 nor'easter.

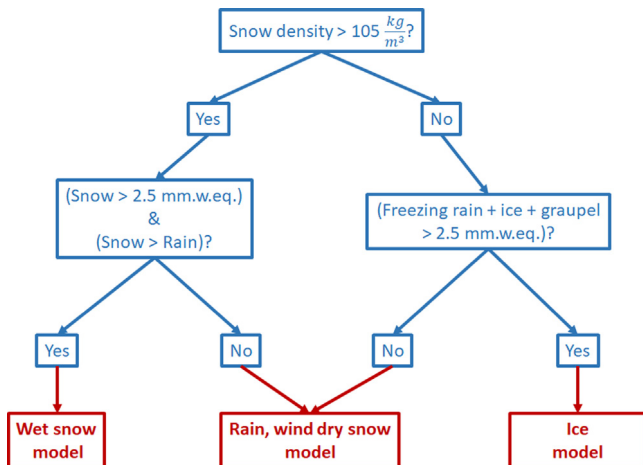


Fig. 4. Algorithm for precipitation type classification.

Based on this algorithm, all grid cells with moderate freezing rain (2b) are classified as ice; grid cells associated with the primary mode of the sleet distribution (2c), co-occurring with average-density snowfall, are classified as ice too; and grid cells associated with the secondary mode are classified as wet snow, since almost all of them co-occur with moderate amounts of heavy snow (2c).

3.4. Performance evaluation method

The performance evaluation of the ML and GLM outage prediction models was demonstrated, similarly to [53], through leave-one-storm-out cross-validations. This method consists in predicting outages related to a storm using a model trained on all the other storms, which resembles real-time applications of the model. In this study, multiple cross-validations were performed for obtaining the best configuration of each model: tests were performed with and without the grid cell classification module, at 4 km grid spacing and at a town level, and with different amounts

Table 2
Configuration of the ML-based and GLM-based models with the best performance.

Model	ML	GLM
Predictive scale	4 km	Town
Weather classification	Yes	No
Variable list	Full	Selected

of variables used as model predictors. The best performing configuration for each model is reported in Section 4, together with their performance.

3.5. Error metrics

The error metrics used for evaluating model performance were:

- the *median absolute percentage error* (MdAPE), for measuring the maximum percentage error committed 50 percent of the times;
- the *mean absolute percentage error* (MAPE), for measuring the average model accuracy;
- the *Nash–Sutcliffe Efficiency* (NSE; [90]), for determining the model efficiency for a number n of predictions P and observations O :

$$NSE = 1 - \left[\frac{\sum_{k=1}^n (P_k - O_k)^2}{\sum_{k=1}^n (O_k - \bar{O})^2} \right] \quad (2)$$

4. Results and discussion

We performed cross-validation experiments for the 54 analyzed outage events using GLM and ML models in different configurations: at 4km and at the town scale, with and without the weather classification algorithm presented in Fig. 4, with all variables listed in Appendix B and through variable selection. The best configuration for both ML models and GLM was found by comparing the different cross-validation results, giving primary importance to the NSE, and in case of similar results, to the MdAPE and MAPE [53].

The best performing ML-based system was the one at 4 km grid spacing, with weather classification at the grid level that used the full list of variables (Table 2). On the other hand, the GLM performance increased when data were aggregated at town level, without precipitation type classification, and with a more restricted list of variables (Table 2). These elements can be explained as follows:

- for the GLM model, the dataset drastically shrinks after aggregation to the town level, therefore a further classification would reduce the amount of data (especially for wet snow) to a level at which it would be difficult for GLM to extract useful information. Moreover, over the territory of a town, precipitation type varies (e.g. between mountains and valleys) and the most common precipitation type is not representative for the entire town;
- a model is easily overfitted when the dataset size is small and the number of variables is large. This is the case for the GLM with the full list of variables. After decreasing the number of variables, the model becomes well trained and there is a resulting performance boost in the cross-validation experiments;
- the more sophisticated ML model instead is able to better navigate into the non-linearities of a larger dataset at a finer scale. However, it is beneficial to guide such model using the algorithm shown in Fig. 4, which is based on physical evidence or statistical findings.

In the following sections the variable importance for the best performing models is presented, together with models performance at service territory and at town levels.

4.1. Variable importance

Since variable importance is computed through different method across the different models, normalized importances are computed and presented in Fig. 5. For the machine learning models, variable importance plots for each of the three precipitation types are provided by the inclusion proportions method [88] for BART and the permutation importance [86] for RF. For the GLM a single importance plot for the variables resulting from a stepwise regression [93] is presented, since the best performing model is not trained separately for different precipitation types. The elimination selection [94] for the variables is based on the residual sum of squares.

All the models agreed on the two most important variables: assets on the territory (sumIso) and LAI. The first finding is in agreement with previous studies [42], and is a direct consequence of the proportionality between the number of failures and the number of assets [95]. The second finding is connected to outage causality: trees are the primary cause of failures and the LAI allows the models to understand where wintry precipitation can accumulate on trees, increasing the risk of power outages.

Beyond this agreement on the high importance of these two variables, other relevant findings can be summarized as follows:

- both machine learning models agreed in assigning a high importance to freezing rain and much lower to ice in the freezing rain and sleet model. Freezing rain in fact accumulates on trees and infrastructure, while sleet does not. The importance of these variables is much lower in the GLM, since these variables are used for describing only on a small fraction of the dataset;
- the importance of coniferous and deciduous forest, and of the product between LAI and snow is high for the wet snow ML models, since those variables are related to vegetation caused outages;
- the high importance of snow density across all the models confirms its primary role, discussed in Section 2.4;
- maximum gusts are very important for all the models, since the interaction between gusts, trees and infrastructure, already discussed in [53], is a primary cause of outages. Gust is a parameter, beyond ice accumulation, used in the [85] ice accumulation index, and the high importance in the ice models developed in this work confirms its role.

4.2. Models performances

Differences in the principles behind the ML- and GLM-based systems reflect in intrinsic performance differences that can be rationally explained.

Fig. 6 provides a side-to-side comparison of the performances of the two models described in Table 2, in terms of storm total outages. In each scatterplot, predicted versus actual outages are reported for each of the 54 storms, color coded according to the dominant precipitation type.

Since both RF and BART models find patterns by classifying the data, it is not possible for the ML-based outage model to predict the most extreme event (the October 2011 nor'easter), because data of that magnitude are not present in the rest of the dataset. The GLM instead fits functions across variables, that can be potentially extended beyond the dynamic range of the training dataset. For the 2011 storm, the GLM was able to extrapolate a value close to the actual, by using a training dataset in which the

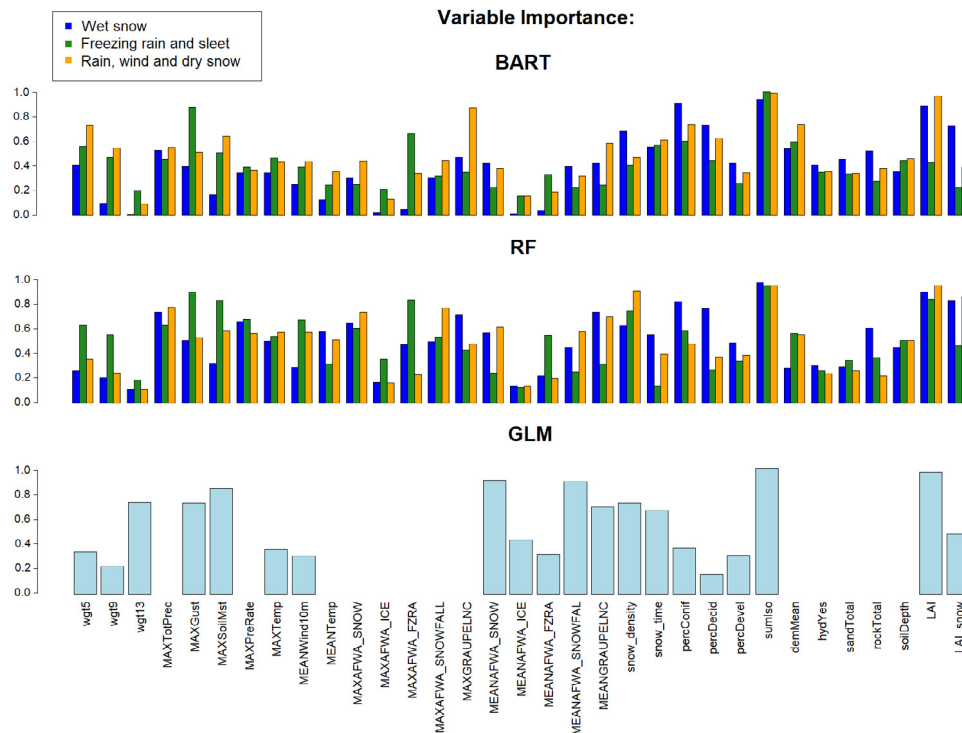


Fig. 5. Variable importance for the BART, RF, and GLM.

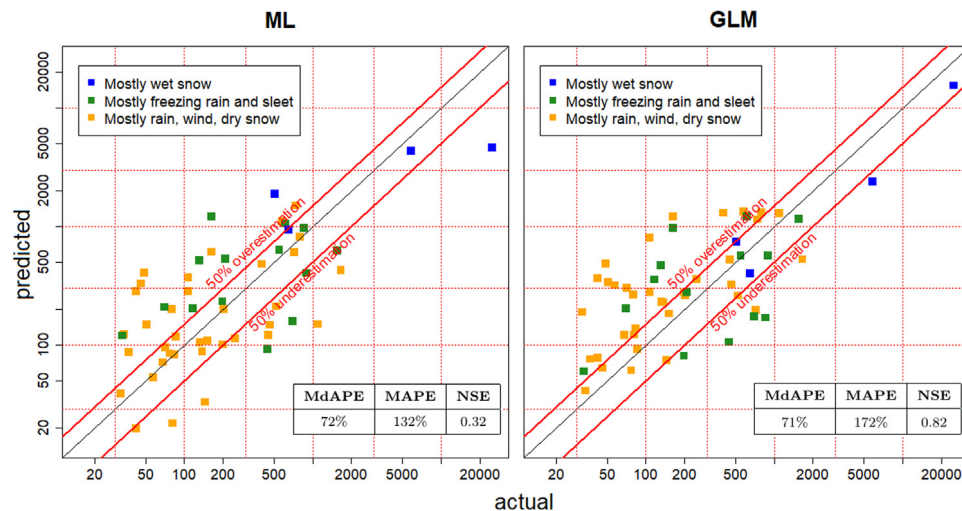


Fig. 6. Scatter plots of total storm outages for the cross-validation results for the ML-based and the GLM-based OPM.

strongest event (March 2018) has a magnitude four times lower than the October nor'easter.

GLM is generally better performing than ML models for medium- and high-impact events (above 300 outages), especially for wet snow events, while the GLM is biased for low-impact events, resulting in an overestimation of rain, wind and dry snow events. The OPM shows less variability for different precipitation types and magnitude for the ML-based model, since each precipitation type has its own unbiased model. The ML-based OPM has better performance than the GLM for predicting the more frequent low-impact events.

These characteristics reflect in the performance statistics (shown on the bottom right corner of Fig. 6): the high NSE of the GLM comes from the good prediction of strong events, while its higher MAPE is due to the very high relative errors in low-impact events prediction. The opposite is true for the ML-based models.

The spatial distribution of the error, provided in Fig. 7, highlights similarities but also differences between the two systems. For both OPMs, consistent overestimations are noted for the northwestern and southeastern Connecticut (areas where the number of assets, hence the expected number of outages, is lower), while underestimations can be found in the densely populated southwestern and central Connecticut. Error magnitude is generally higher for the GLM (darker colors), and lower for the ML (lighter colors), suggesting that the ML-based OPM is able to more accurately capture the spatial distribution of outages.

4.3. The 2011 october nor'easter

On October 29, 2011 a powerful nor'easter characterized by strong winds and heavy snow on fully leaved trees produced a number of outages higher than the outages produced by all the

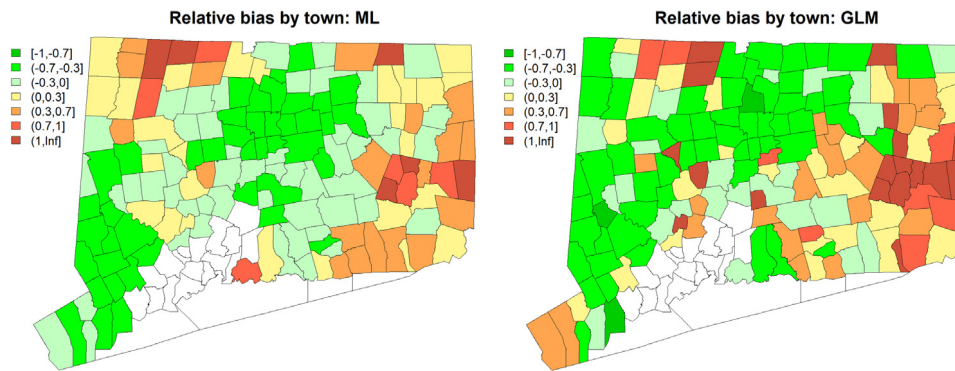


Fig. 7. Relative bias for the ML-based and the GLM-based OPM.

other 53 snowstorms in our database together: due to this storm, 25,019 locations required manual intervention by utility crews in the service area, and power restoration for homes and businesses lasted 11 days. This section aims at analyzing the key factors that led to the catastrophic outcome of this event, and the reasons responsible for a southeastward shift in the spatial distribution of the predicted outages, when compared to the actual distribution.

In Section 2.4 the importance of snow density was highlighted. According to the snow density map derived from WRF for the October 2011 event (Fig. 8a), very wet snow fell in coastal areas in Southern Connecticut and in the densely populated, low-elevation terrains of the Connecticut river valley (the darker vertical band in Central Connecticut). In Central and Southeastern Connecticut, wet snow was associated, according to WRF, to high amounts of snow (20 to 30 mm water equivalent, Fig. 8b). However, the NOAA Regional Snow Analysis (Fig. 8c, available at <https://www.noahrs.noaa.gov/nsa/>) shows that the highest amounts of snow occurred in Central and Northwestern Connecticut, and that snow amounts were lower in Southeastern Connecticut. The maps of predicted (Fig. 8e,f) and actual (Fig. 8e,f) outages closely reflect these differences: (i) the predicted hotspot of high amount of outages in Southeastern Connecticut did not occur because a lower amount of snow fell in those areas, (ii) high amounts of predicted outages in Central and Southwestern Connecticut (mostly visible in the GLM prediction) were correctly captured, since the WRF analysis was consistent with NOAA analysis, (iii) the amount of outages in Northeastern Connecticut was severely underestimated, because the amount of snow predicted by WRF was less than half than the amount of snow displayed in NOAA's map.

This event is emblematic of the enhanced sensitivity of snow and ice storms outage prediction to weather model simulation uncertainty. Beyond the critical role of wind, gusts, and precipitation amount forecasts for rainy extratropical storms and thunderstorms [53], precipitation type and snow density uncertainty adds complexity in winter storms outage prediction and is primary responsible of model inaccuracies.

4.4. Discussion

As noted for the October 2011 nor'easter event, error characteristics of the two systems, along with possible weather forecasting errors, should be taken into consideration when model predictions are used for storm preparedness. Predictions with median errors around 70% may provide a relevant added value for taking preventive measures ahead of storm impacts. It is well known, in fact, that utilities may be underprepared or unprepared (meaning underestimating the impact by one or two orders of magnitude) to the impact of strong winter events, since they

do not have enough information for quantifying their impacts (outages in our dataset span three orders of magnitude).

The two snow and ice OPM systems proposed in this work allow for an identification of the very strong events and prediction of most of the medium and low-impact events. Inaccuracies in weather predictions propagate and magnify in outage prediction, due to non-linear interactions between weather and infrastructure. In outage prediction related to winter storms, such non-linearities are enhanced, and lower performance of winter models compared to rainstorms and thunderstorms performance [53] are expected due to added uncertainties. As shown in Fig. 3, an increase of just 5 kg/m³, between 105 and 110 kg/m³ produces an average 10-fold increase of the number of outages, and such errors are not uncommon, since they are associated with small errors in weather model temperature.

It is also important to note that this version of the models produces a spatial distribution of outages that is smoother than the actual distribution. A direct consequence of this characteristic is that model predictions should be enhanced, meaning that an even higher number of outages should be expected, on average, in locations where the model already predicts a high number of outages, while, on the other hand, a lower number of outages should be expected in rural areas.

5. Concluding remarks

This study presented two systems for the prediction of power outages related to snow and ice storms, and evaluated their performance during 54 snow and ice events occurred between 2005 and 2018. The machine learning based system used 33 variables, a precipitation type classifier, and an optimization module, to predict outages on a 4 km regular grid, using a random forest and a Bayesian additive regression tree model. The system based on a generalized linear model used 20 variables to predict outages at a town level.

From the analysis of the variable importance of the two systems it resulted that assets, leaf area index and snow density have the primary importance. These three variables describe the quintessential relationships between trees, weather and infrastructure. In the heavy snow model additional variables related to vegetation were important, while freezing rain amount and gusts were very important for the ice model.

The GLM is able to more accurately predict extreme events (outside the dynamic range of the training dataset), while the ML models would underestimate the most severe nor'easter. ML models have better performance than the GLM for the prediction of events within the dynamic range of the training dataset, and have lower errors in terms of the spatial distribution of outages. An analysis of the reasons of the spatial errors for the devastating 2011 October nor'easter identified the misrepresentation

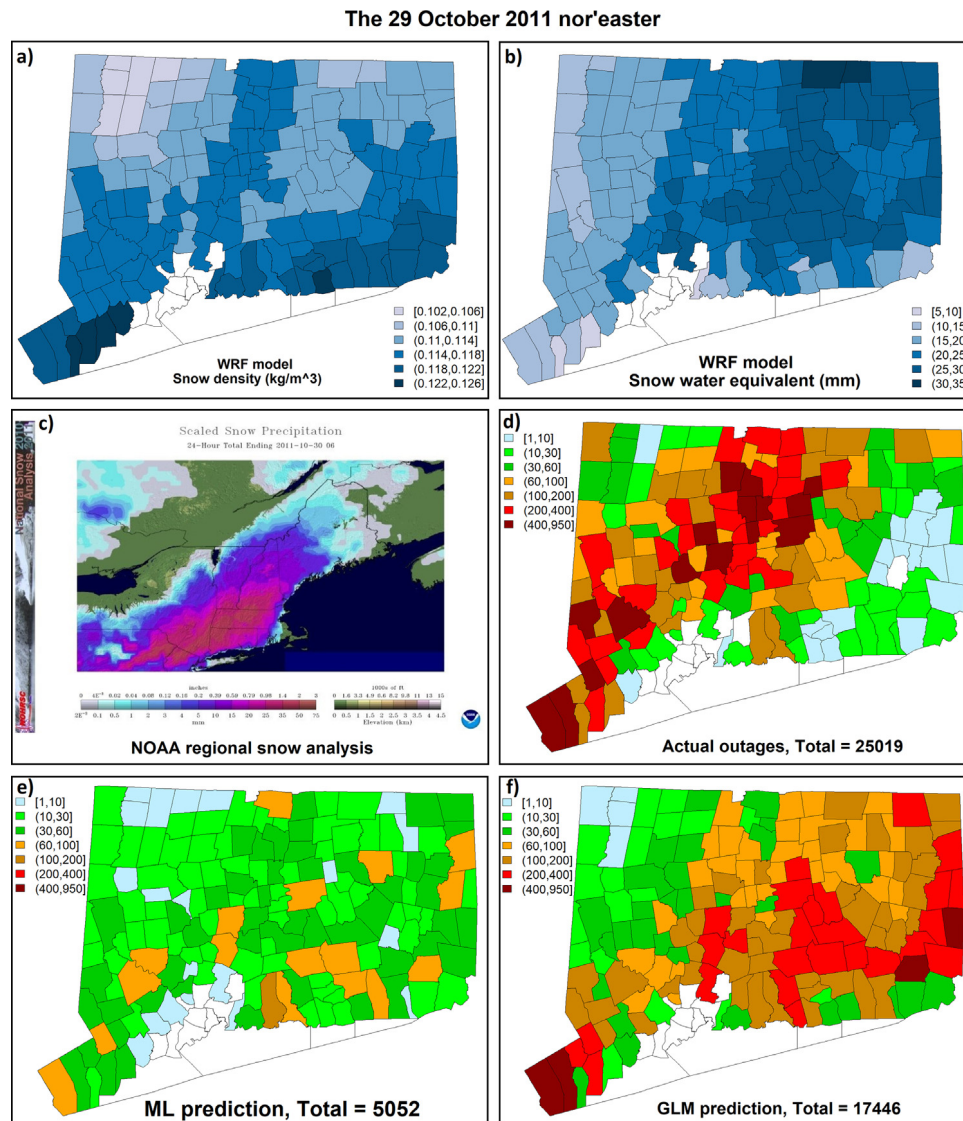


Fig. 8. The October 29, 2011 nor'easter: (a) snow density and (b) snow water equivalent analysis from WRF; (c) snow water equivalent analysis from NOAA; (d) actual outages; outages predicted using (e) Machine Learning models, and (f) the Generalized Linear Model.

of weather phenomena being the primary cause for a spatial mismatch between predicted and actual outages.

Both GLM and ML models are currently used in real-time for predicting winter storms power outages on the Eversource-Connecticut electric distribution grid. The utility takes into account of these predictions for deciding about crews and equipment allocation in advance of winter events.

Future improvements of these models include the study of squared effects and interactions between variables in the GLM and the exploration of a possible blending of the two predictive systems, for creating a unified outage prediction model for snow and ice storms, valid for predicting extreme winter events and also for capturing the spatial distribution of outages in low-impact events.

Declaration of competing interest

The authors partially uses classified datasets of the electric grid, provided by Eversource Energy. We have full access to all of the data in this study and we take complete responsibility for the integrity of the data and the accuracy of the data analysis. Emmanouil Anagnostou, Diego Cerrai, and Peter Watson hold stock in ACW Analytics.

CRediT authorship contribution statement

Diego Cerrai: Methodology, Software, Formal analysis, Investigation, Validation, Writing - original draft, Visualization. **Marika Koukoulou:** Software, Formal analysis, Data curation. **Peter Watson:** Software, Formal analysis, Data curation. **Emmanouil N. Anagnostou:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing - review & editing.

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Appendix A. Storm list

See [Table A.1](#).

Appendix B. Variable description

See [Table B.2](#).

Table A.1

List of snow and ice storms constituting the historical dataset.

Snow and ice storms					
Date	Outages	Date	Outages	Date	Outages
2005-12-16	131	2011-02-02	199	2016-04-03	740
2006-01-02	549	2011-10-29	25019	2016-12-17	107
2006-01-14	1560	2012-11-07	800	2017-01-23	723
2007-03-02	250	2012-12-28	132	2017-02-09	204
2007-12-02	467	2013-02-08	1097	2017-02-13	523
2008-01-13	652	2013-03-08	401	2017-03-10	71
2008-02-12	696	2014-02-05	33	2017-03-13	209
2008-12-10	615	2014-02-13	117	2017-12-09	81
2008-12-16	57	2014-11-26	507	2017-12-23	138
2008-12-30	454	2015-01-18	51	2018-01-04	164
2009-01-07	896	2015-01-26	46	2018-01-16	42
2009-01-10	34	2015-02-15	86	2018-01-30	37
2009-01-28	77	2016-01-12	84	2018-02-01	80
2010-02-23	163	2016-01-17	68	2018-02-07	70
2010-12-26	857	2016-01-23	42	2018-02-17	200
2011-01-12	152	2016-02-04	1663	2018-02-22	49
2011-01-18	444	2016-02-13	145	2018-03-07	5841
2011-01-26	108	2016-03-20	32	2018-03-13	577

Table B.2

Description of the predictor variables used in the dataset.

Variable [Units]	Description
Wgt5, 9, 13 [h]	Duration of wind at 10 m above 5, 9, 13 m/s
MaxTotPrec [mm]	Total precipitation
MaxGust [m/s]	Maximum wind gusts at 10 m
MaxSoilMst [mm mm ⁻¹]	Maximum soil moisture
MaxPreRate [mm h ⁻¹]	Maximum precipitation rate
MaxTemp [K]	Maximum temperature
MeanWind10 m [m s ⁻¹]	Mean wind at 10 m
MeanTemp [K]	Mean temperature during 4 hrs of max wind
MaxGraupelInc [mm]	Total graupel accumulation
MeanGraupelInc [mm]	Graupel accumulation during 4 hrs of max wind
MaxAFWA_snow [mm]	Total liquid equivalent snow
MaxAFWA_ice [mm]	Total ice pellet accumulation
MaxAFWA_fzra [mm]	Total freezing rain accumulation
MaxAFWA_snowfall [mm]	Total snow accumulation
MeanAFWA_snow [mm]	Liquid equivalent snow during 4 hrs of max wind
MeanAFWA_ice [mm]	Ice pellet accumulation during 4 hrs of max wind
MeanAFWA_fzra [mm]	Freezing rain accumulation during 4 hrs of max wind
MeanAFWA_snowfall [mm]	Snow accumulation during 4 hrs of max wind
Snow_density [kg m ⁻³]	Snow density
Snow_time [-]	ratio between snow during 4 hrs of max wind and total snow
LAI_snow [%]	Product between LAI and total snow
PercDecid [%]	Percentage of deciduous forest
PercConif [%]	Percentage of coniferous forest
PercDevel [%]	Percentage of developed area
DemMean [m]	Mean elevation
HydrYes [%]	Percentage of hydric soil
SandTotal [%]	Percent of sand in soil
RockTotal [%]	Percent of rock in soil
Soil depth [m]	Mean soil depth
SumIso [-]	Total number of isolating devices
LAI [m ² m ⁻²]	Leaf area index

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